

A Modular, Self-Healing, and Evolvable Neural Architecture Inspired by Biological Systems

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Outline

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Biological Inspiration

- Motivation: Mimic the adaptability, self-healing, and evolution of biological organisms.
- Cells act as autonomous learning units.
- DNA-like encoding tracks traits such as learning rate, activation mix, dropout, etc.
- Supports adaptability under environmental stress like noise, adversarial data, or failure.

Modular Cell Architecture

- Each **Neural Cell** is an independent module.
- Encodes behavior using a probabilistic trait map:

$$\text{DNA} = \{\text{activation_mix}, \text{learning_rate}, \text{dropout}, \dots\}$$

- Communication occurs via superposition and entanglement (shared memory/state).
- Mutation trait selection inspired by evolutionary biology and Bayesian optimization.

Cellular Self-Healing

- Damaged or poorly performing cells are isolated.
- Their encoded DNA is evolved using:
 - Loss score differences
 - Functional divergence (measured via output drift)
 - Trait success rate (recorded over epochs)
- Repaired cells are reintroduced with enhanced encoding.

Incorporating Superposition and Entanglement

- **Superposition:** Each cell can maintain a mixed state of multiple behaviors.
- **Entanglement:** Correlated learning across distant cells.
- Increases convergence speed and robustness of learning.
- Shared loss landscapes inform DNA updates globally.

Evolution-Driven Training

- Each cell undergoes localized backpropagation.
- Loss is stored and analyzed by the DNA controller.
- Cells evolve based on:
 - Local performance
 - Global influence (using attention-like weighting)
- Population control ensures optimal cell number.

Core Data Structures

- Graphs: Cell connectivity and interaction modeling.
- Heaps: Priority-based cell regeneration.
- Queues: Temporal buffering of training data.
- Hash maps: Efficient DNA encoding and trait lookup.

Directions Forward

- Integration with neuromorphic hardware.
- Real-time cell mutation guided by environmental sensing.
- Federated multi-agent evolution (cross-device DNA).
- Further exploration of transformer-based topologies in cell layers.

Conclusion

- A biologically inspired architecture with resilience, adaptability, and evolution.
- Combines principles of neural networks, quantum theory, and biology.
- Introduced a robust, modular training and self-healing framework.

Thank You

Questions?